Gas Report

Project 7 — NFT Event Ticketing (ERC-721)

hardhat-gas-reporter v1.0.10

Environment Configuration

Command: `REPORT_GAS=true npx hardhat test`

Solc: 0.8.28 **Optimizer:** enabled **Runs:** 200

viaIR: true

Network: hardhat (chainId 31337)

Block gas limit: 60,000,000

Gas price (ref): 1 gwei (local; mainnet figures derived in §D)

Gas Usage Table

Solc version: 0.8.28 **Optimizer enabled:** true **Runs:** 200 **Block limit:** 60,000,000

Contract	Method	Min	Max	Avg	# calls
EventTicketNFT	createEvent	240,951	357,606	296,739	21
EventTicketNFT	buyTicket	180,568	285,980	250,776	28
EventTicketNFT	buyMultipleTickets	—	—	470,948	1
EventTicketNFT	buyResaleTicket	—	—	98,918	1
EventTicketNFT	listForResale	154,319	171,419	169,282	8
EventTicketNFT	cancelResaleListing	—	—	45,388	1
EventTicketNFT	addTicketsToSection	—	—	44,255	2
EventTicketNFT	updateEvent	39,230	47,185	44,533	3
EventTicketNFT	invalidateTicket	30,688	46,160	41,003	3
EventTicketNFT	cancelEvent	—	—	30,676	4
Deployments					
EventTicketNFT		—	—	3,539,138	5.9 % of block limit

Headline Numbers

(the three rubric-relevant entry points)

→ `createEvent` avg 296,739 gas

→ `buyTicket` avg 250,776 gas

→ `buyResaleTicket` avg 98,918 gas

→ **Deployment** 3,539,138 gas (5.9 % of block limit)

Optimisation Delta on `createEvent`

(full write-up: project-evaluation.md §D)

BEFORE (on-chain strings: name, description, category, banner, sec. labels)
avg = 389,440 gas   min = 313,285   max = 480,147

AFTER (single IPFS CID stored on-chain; all human prose moved off-chain)
avg = 296,739 gas   min = 240,951   max = 357,606

SAVED = ∼92,701 gas / event (-23.8 % avg ; -25.5 % at the worst case)

What this looks like in ETH / INR

(createEvent only)
ETH = \$3,000   \$1 = 83

Gas price (gwei)	Saved ETH/event	Saved USD/event	Saved INR/event
30	0.00278 ETH	∼\$8.34	∼692
60	0.00556 ETH	∼\$16.69	∼1,385
100	0.00927 ETH	∼\$27.81	∼2,308

For an organizer publishing 100 events, this is a ∼70k–230k saving with zero functional regression - the JSON metadata is unchanged, only its storage location moved from the EVM to IPFS.

Reproduce

```
npm install
npm run gas-report          # prints the table above to stdout
npm run gas-report:file     # also writes reports/gas-report.txt
```